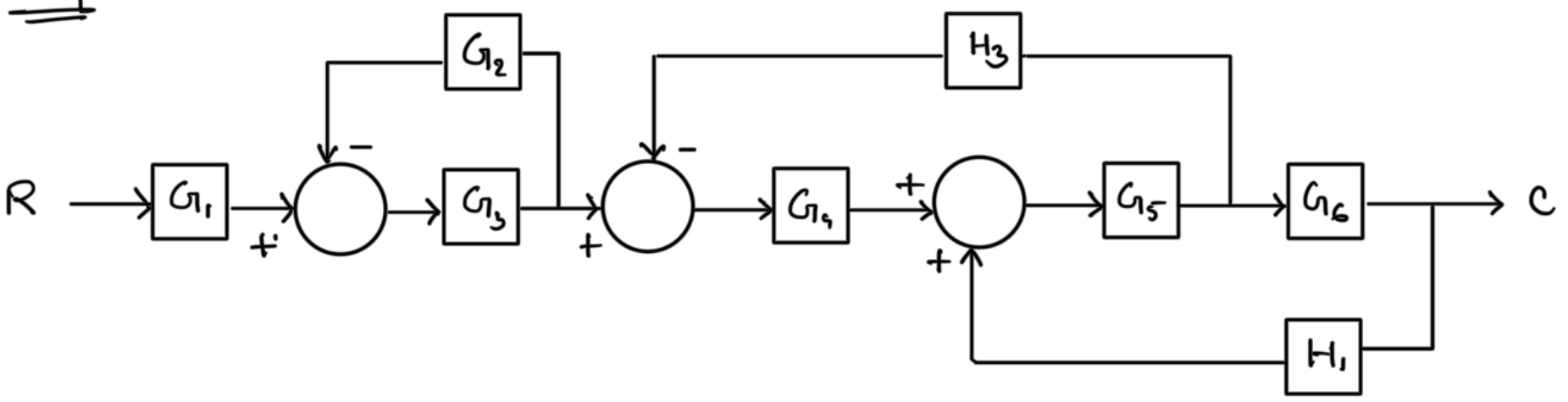
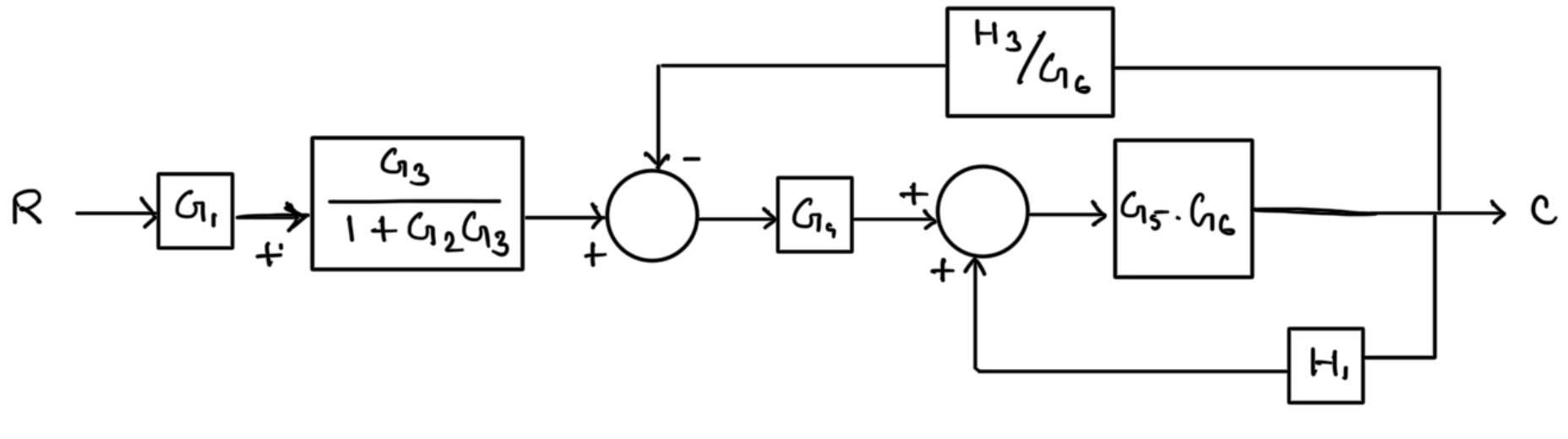
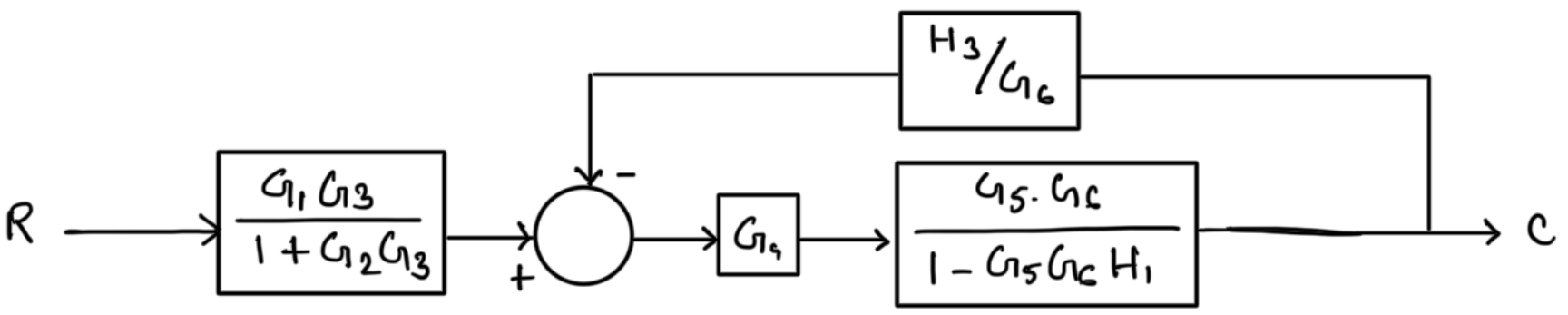
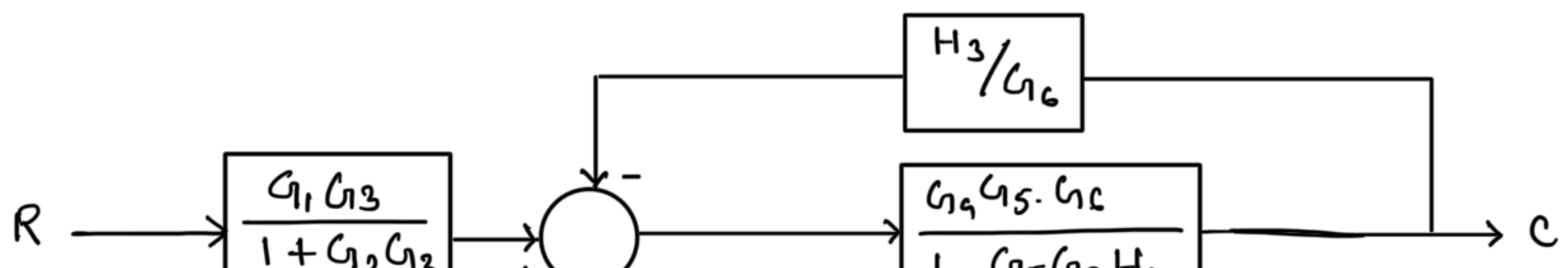
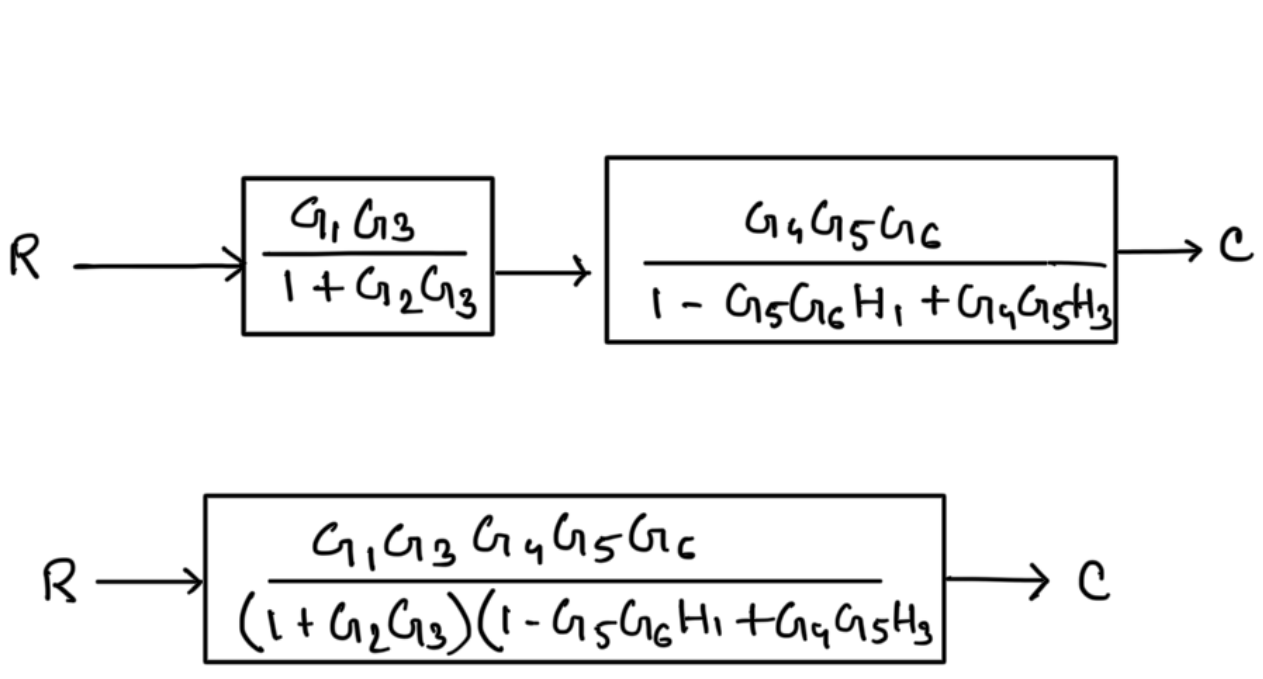
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$$\frac{\frac{G_4G_5G_6}{1-G_5G_6H_1}}{1+\frac{G_4G_5G_6}{1-G_5G_6H_1}\times\frac{H_3}{G_6}}$$

$$= \frac{G_4G_5G_6}{1-G_5G_6H_1+G_4G_5H_3}$$

2| a| $K = 0.8$, $P = 10$

$\therefore K_p = 0.45 \times 0.8$ $\quad K_i = 1.2/10$

$= 0.36$ $\quad = 0.12$

b) desired speed = 35

Step	Current Speed	Error	output	New Speed
1	20	15	$= 0.36 \times 15 + 0.12 \times 15 = 7.2$	27.2
2	27.2	7.8	$= 0.36 \times 7.8 + 0.12 \times (15 + 7.8) = 5.5$	32.7
3	32.7	2.3	$= 0.36 \times 2.3 + 0.12 (15 + 7.8 + 2.3) = 3.8$	36.5
4	36.5	−1.5	$= 0.36 \times (-1.5) + 0.12 (15 + 7.8 + 2.3 - 1.5) = 2.292$	38.792